

Taming implies compatibility on four-manifolds

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Abstract

We prove that every smooth almost complex structure on a closed four-manifold which is tamed by a symplectic form is compatible with a symplectic form. This gives a positive solution to Donaldson's tamed-to-compatible conjecture. We also characterize the compatible cone by strict intersection positivity on the closure of the invariant part of the taming cone.

1 Introduction

Let X be a closed smooth four-manifold and let J be a smooth almost complex structure on X . A smooth real two-form ω is symplectic if it is nondegenerate and closed, meaning $d\omega = 0$. A real two-form ω *tames* J if $\omega(v, Jv) > 0$ for every nonzero tangent vector v . It is *compatible* with J if, in addition, $\omega(Ju, Jv) = \omega(u, v)$. A taming form is automatically nondegenerate; thus a closed taming form is symplectic.

Donaldson asked whether the existence of a taming symplectic form on a closed four-manifold implies the existence of a compatible one [2, Section 5.2, Question 2]. This existence question is distinct from his a priori estimate conjecture for a prescribed-volume equation. The elementary projection $\omega \mapsto \frac{1}{2}(\omega + \omega(J \cdot, J \cdot))$ shows the obstacle: it preserves positivity on complex lines but generally loses closedness. Our main result answers the existence question affirmatively.

Theorem 1.1. *Suppose that a smooth almost complex structure J on a closed connected smooth real four-manifold X is tamed by a symplectic form. Then there is a smooth symplectic form compatible with J .*

The almost complex structure in Theorem 1.1 is the given one. The conclusion imposes no condition on the cohomology class of the compatible form. In particular, it does not assert that an arbitrary taming class contains a compatible representative.

There is also an intrinsic description of the compatible cone in terms of the taming cone and the intersection pairing.

Theorem 1.2 (Taming and compatible cones). *Let X, J satisfy the hypotheses of Theorem 1.1, and give X the orientation induced by J . Let $\mathcal{K}_J^t, \mathcal{K}_J^c \subset H^2(X; \mathbb{R})$ be the classes represented by taming and compatible symplectic forms, respectively. Let H_J^- be the subspace represented by smooth closed real anti-invariant two-forms. Set*

$$\mathcal{V} = (H_J^-)^\perp, \quad \mathcal{C} = \mathcal{K}_J^t \cap \mathcal{V},$$

where orthogonality is for the intersection pairing. Then

$$\mathcal{K}_J^t = \mathcal{C} + H_J^-, \tag{1.1}$$

$$\mathcal{K}_J^c = \{a \in \mathcal{C} : a \cdot y > 0 \text{ for every } 0 \neq y \in \overline{\mathcal{C}}\}. \tag{1.2}$$

The closure is taken in $\mathcal{V}$, and the sum in (1.1) is a Minkowski sum.

Corollary 1.3. *Under the same hypotheses, write $h_J^- = \dim H_J^-$ and let $b_2^+(X)$ be the positive index of the intersection form. If $h_J^- = b_2^+(X) - 1$, then*

$$\mathcal{K}_J^t = \mathcal{K}_J^c + H_J^-.$$

In particular, if $b_2^+(X) = 1$, then $\mathcal{K}_J^t = \mathcal{K}_J^c$: every taming class has a smooth compatible symplectic representative.

The proofs are given in Section 8. They use separation in a prescribed cohomology class and the current estimates below. The inclusion $\mathcal{K}_J^c + H_J^- \subseteq \mathcal{K}_J^t$ can be strict when $h_J^- < b_2^+ - 1$, even if $H_J^- = 0$; Example 8.6 gives an explicit four-torus.

The integrable case belongs to the theory of compact complex surfaces. Buchdahl [1, Theorem 11] and Lamari [9, Corollary 5.7] gave classification-free proofs that even first Betti number implies the existence of a Kähler metric. Lamari's positive exact current on a surface with odd first Betti number also obstructs a taming form [9, Theorem 6.1]. Li and Zhang proved the equivalence between the existence of taming and compatible symplectic forms on complex surfaces [12, Theorem 1.2].

The pseudoholomorphic-curve approach has a different starting point. For structures on $\mathbb{CP}^2$ tamed by the standard symplectic form, Gromov constructed a compatible form by averaging currents of pseudoholomorphic lines [6, Section 2.4.A']. Taubes developed integration over curve moduli spaces to obtain compatibility for a residual subset of the smooth almost complex structures tamed by a fixed symplectic form when $b_2^+ = 1$ [21, 22]. Theorem 1 of [21] also states preservation of the taming form's cohomology class when that class is rational. Li and Zhang extended this approach to every tamed structure on $S^2 \times S^2$ and $\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$, and to further rational manifolds under hypotheses on embedded pseudoholomorphic spheres [13, Theorems 1.2 and 1.3].

An invariant two-form satisfies $\alpha(Ju, Jv) = \alpha(u, v)$; an anti-invariant one satisfies $\alpha(Ju, Jv) = -\alpha(u, v)$. Write $h_J^- = \dim H_J^-$, where H_J^- is the anti-invariant cohomology space defined in Theorem 1.2. The condition $h_J^- = b_2^+ - 1$ occurs in Buchdahl-type approaches [20, 14], in the compatibility criteria of Wang, Wang and Zhu [24, Theorems 4.3 and 5.1], and in the taming corollary of Wang, Zhang, Zheng and Zhu [25, Corollary 1.3]. The latter paper's generalized Monge–Ampère theorem starts with an almost Kähler structure. Li and Ning's recent study of spaces of Kähler and holomorphically tamed forms allows the integrable complex structure to vary [11].

The current approach has a classical cone-duality background. Sullivan related closed forms positive on a field of tangent directions to the corresponding structure currents [19]. Harvey and Lawson characterized Kähler manifolds through positive currents [7]; Li and Zhang developed the almost-complex formulation used to compare taming and compatibility [12, Section 3]. In the argument below, separation gives a positive functional vanishing on closed invariant forms. An explicit elliptic lift and quantitative current estimates connect this obstruction to the existence of a taming form.

The argument and its main estimate

Fix an auxiliary J -Hermitian metric. If no compatible form exists, separation produces a nonzero positive current P that annihilates closed invariant forms. An elliptic correction gives $T = P + Q$, which annihilates every closed two-form, with Q initially an anti-invariant distribution. Represent P as a positive weighted average of unit complex-line bivectors, and let μ be the resulting measure on X , its trace measure. Corrected radial test functions give $\mu(B_r(x)) \leq Cr^2$. Regularization by a

finite power of the Hodge resolvent then proves $Q \in L^2$ and an angular estimate: over pairs of complex lines in this representation whose base points are less than r apart, the integral of their squared difference, after parallel transport, is $O(r^4)$.

The first regularized pairing only bounds $\|Q\|_2$, because the negative error in this pairing is only uniformly bounded. To remove that error, we distinguish the points where P has positive two-dimensional density. Write

$$\theta(x) = \lim_{r \downarrow 0} r^{-2} \mu(B_r(x)), \quad E = \{x : \theta(x) > 0\}.$$

Existence of the limit is part of the argument. The principal step, Theorem 5.1, proves that the restriction $P_E = \mathbf{1}_E P$ is closed. It applies to any closed current $P + Q$ with an anti-invariant L^2 correction and the stated mass and angular bounds; annihilation of all closed forms is not needed for this step.

The reason the angular estimate suffices is a balance between two errors. Planar tangent measures show that, over the same nearby pairs, the squared normal component of the displacement between base points, relative to the first complex line and divided by r^2 , has integral $o(r^2)$. Closedness of $P + Q$ trades tangential derivatives of a smooth density cutoff for a product of angular and transverse errors. The derivative and normalization of the cutoff contribute r^{-3} . Cauchy–Schwarz therefore bounds the decisive boundary term by

$$r^{-3} O(r^2) o(r) = o(1),$$

where the two factors are the square roots of the angular and transverse moments. The L^2 bound controls the correction’s cutoff error separately. Together these estimates prove closedness of the density restriction without requiring a quantitative rate for tangent measure convergence.

Once P_E is closed, the residual $T - P_E = \mathbf{1}_{X \setminus E} P + Q$ is closed as well. Its positive summand is concentrated on zero-density centers, where the negative error in the pairing does tend to zero. Pairing this residual with T , and using the vanishing of the mixed terms, forces $Q = 0$. The resulting nonzero positive current $P = T$ cannot annihilate a closed taming form.

The relative projection estimate in Lemma 4.4, for fixed smooth orthogonal bundle projections, and the closedness statement in Theorem 5.1 may be useful separately. The regularized pairing estimates use the four-dimensional fact that anti-invariant two-forms are self-dual. Sections 2 and 3 construct the separating current and prove its mass bound. Section 4 establishes the regularization and angular estimates. Section 5 proves density existence and the planar tangent description; Section 6 uses them to control cutoff boundaries and prove closedness of P_E . Section 7 completes the existence argument. Section 8 develops the prescribed-class separation and proves the cone description.

A triple of closed two-forms on an oriented four-manifold is *hypersymplectic* if every nonzero real linear combination is symplectic and induces the given orientation. The companion [18, Theorem 1.1] combines the current estimates above with additional prescribed-class and deformation arguments to deform hypersymplectic triples normalized by $\int_X \omega_i \wedge \omega_j = \delta_{ij}$ on closed connected oriented smooth four-manifolds to hyperkähler triples, while fixing each class $[\omega_i] \in H^2(X; \mathbb{R})$.

2 A current obtained by separation

We first construct the current that would obstruct compatibility: a positive current and an anti-invariant correction whose sum annihilates every closed two-form.

Give X its complex orientation. Fix a smooth J -Hermitian metric g , and write

$$F(u, v) = g(Ju, v), \quad (R\alpha)(u, v) = \frac{1}{2}(\alpha(u, v) - \alpha(Ju, Jv)).$$

Thus R is the orthogonal projection onto the anti-invariant two-forms, and $1 - R$ projects onto the invariant two-forms. All forms and currents are real.

We identify bivectors with two-forms using g . Let $\mathcal{C} \rightarrow X$ be the bundle of unit complex line bivectors

$$\mathcal{C}_x = \{v \wedge Jv : |v|_g = 1\}.$$

For $\ell \in \mathcal{C}_x$, let $L_\ell \subset T_x X$ be its oriented plane. The following pointwise identities will be used:

$$\text{im } R \subset \Lambda_g^+, \quad *F = F, \quad |F|^2 = 2, \quad |\ell| = 1, \quad \ell^+ = F/2, \quad F(\ell) = 1. \quad (2.1)$$

They follow by taking a unitary frame (v, Jv, w, Jw) . In that frame F corresponds to $v \wedge Jv + w \wedge Jw$, and the anti-invariant subspace is spanned by

$$v \wedge w - Jv \wedge Jw, \quad Jv \wedge w + v \wedge Jw.$$

A two-current acts on smooth two-forms. It is *closed* if it annihilates exact two-forms. Under the L^2 identification with distributional two-forms, this means $d^*T = 0$, or equivalently $d(*T) = 0$. Multiplication and orthogonal projections on distributional forms act on currents by duality.

Lemma 2.1 (A closed lift). *There is a real order-zero pseudodifferential operator*

$$B : C^\infty(\text{im } R) \longrightarrow \Omega^2(X)$$

such that $dB = 0$ and $RB = \text{id}$.

Proof. Consider $\mathcal{L} = Rdd^*$ on sections of $\text{im } R$. If $\xi \neq 0$, the projection of $\xi \wedge \iota_{\xi^\#}$ from self-dual two-forms back to self-dual two-forms is $|\xi|^2/2$ times the identity. By (2.1), the principal symbol of $\mathcal{L}$ on $\text{im } R$ is therefore the same scalar. In particular $\mathcal{L}$ is elliptic. For anti-invariant smooth forms a, b ,

$$\langle \mathcal{L}a, b \rangle_{L^2} = \langle d^*a, d^*b \rangle_{L^2}.$$

It is nonnegative and self-adjoint. Every element of its kernel is smooth and coclosed; being self-dual, it is also closed.

Let H be the orthogonal projection onto this kernel and let G be the inverse of $\mathcal{L}$ on its orthogonal complement, extended by zero on the kernel. The elliptic generalized-inverse theorem gives G of order -2 and H a smoothing finite-rank operator; see [17, Proposition 6.4 and Theorem 6.3]. Set

$$B = dd^*G + H.$$

Then $dB = 0$, while $RB = \mathcal{L}G + H = \text{id}$. □

The projected elliptic operator used here also appears in Lejmi's proof of local compatibility in dimension four [10, proof of Theorem 1]. Drăghici and Zhang's exact-form result gives the corresponding global linear correction phenomenon [3, Proposition 3.1]. The present formula specifies the closed lift and its mapping properties; positivity remains a separate problem.

Proposition 2.2. *If J admits no compatible symplectic form, there are currents P, Q, T and a finite positive measure λ on $\mathcal{C}$ such that*

$$P(\alpha) = \int_{\mathcal{C}} \alpha_x(\ell) d\lambda(x, \ell), \quad \lambda(\mathcal{C}) = 1, \quad (2.2)$$

$$T = P + Q, \quad RP = 0, \quad RQ = Q, \quad *Q = Q, \quad (2.3)$$

$$T(\alpha) = 0 \quad \text{for every smooth closed two-form } \alpha. \quad (2.4)$$

The current P annihilates every closed invariant two-form. The projection μ of λ to X is the trace measure of P : for every smooth function b , $P(bF) = \int_X b d\mu$. Each of P, Q, T belongs to H^{-3} .

Proof. In the Fréchet space of smooth invariant two-forms, the cone of positive definite forms is open and convex. It is disjoint from the linear subspace of closed invariant forms by hypothesis. Hahn–Banach separation gives a nonzero continuous functional nonnegative on the cone and zero on that subspace. Extend it to all two-forms by composing with $1 - R$, and call it P . This is the positive-current separation framework of Sullivan and Harvey–Lawson [19, 7]; for its almost-complex formulation, see [12, Section 3].

Positivity extends to semidefinite invariant forms by adding εF and taking $\varepsilon \downarrow 0$. Since $C\|\alpha\|_{\infty}F \pm (1 - R)\alpha$ are semidefinite for a fixed norm-comparison constant,

$$|P(\alpha)| \leq C\|\alpha\|_{\infty}P(F).$$

Thus P has order zero, and $P(F) > 0$ because $P \neq 0$. Scale P so that $P(F) = 1$.

For completeness, the measure representation can be obtained in local unitary frames. Invariant two-forms identify with Hermitian symmetric forms by $\alpha \mapsto \alpha(\cdot, J\cdot)$. The positive functional has a positive Hermitian matrix of coefficient measures. Its trace is a positive measure μ ; the coefficient measures are absolutely continuous with respect to μ . Their Radon–Nikodym density is a positive semidefinite Hermitian matrix of trace one, or equivalently a convex combination of bivectors in $\mathcal{C}_x$. Measurable spectral decomposition in measurable unitary frames gives λ in (2.2). The identity $F(\ell) = 1$ shows that its projection is μ and that its total mass is one.

Define

$$T(\alpha) = P(\alpha - BR\alpha), \quad Q(\alpha) = -P(BR\alpha).$$

For closed α , the form $\alpha - BR\alpha$ is closed and invariant, proving (2.4). Moreover $RP = 0$ and $RQ = Q$, so $*Q = Q$ by (2.1). In particular T is a closed current.

A finite measure on a compact four-manifold defines an element of H^{-3} , by the Sobolev embedding $H^3 \subset C^0$. The order-zero operator RB^* is bounded on this Sobolev space by the pseudodifferential Sobolev mapping theorem [23, Proposition 0.5.E], so the same conclusion holds for $Q = -RB^*P$ and T . $\square$

Hereafter C , with or without a subscript, denotes a finite constant independent of the small scale parameters. It may change from one occurrence to the next. Constants are uniform over centers in X .

3 Quadratic mass growth

Let P, μ be as in Proposition 2.2. We use the annihilation of closed invariant forms to bound the mass of P in every small ball. Write $B_s(x)$ for a geodesic ball.

Proposition 3.1. *There is a constant C such that*

$$\mu(B_s(x)) \leq Cs^2 \quad (3.1)$$

for all $x \in X$ and $s > 0$.

The proof uses closed invariant test forms

$$Du = (1 - BR)dd_J^c u, \quad d_J^c u = -du \circ J.$$

The Hessian terms of $dd_J^c u$, with J frozen at a point, form an invariant two-form. Consequently Rdd_J^c is a first-order differential operator on functions.

We first spell out the local estimate needed for the order-zero correction. In dimension four, an order-zero pseudodifferential operator has an off-diagonal kernel bounded by $C|y - z|^{-4}$ in local coordinates [23, Section 0.2, Proposition 0.2.B]. Work in a fixed smooth local bundle trivialization. If f is supported in a coordinate ball of radius $a \leq 1$ and its coefficient derivatives satisfy

$$|\partial^j f| \leq C_j M a^{-j}, \quad (3.2)$$

then its local contribution to the operator is bounded by CM . Indeed, write $f(z) = M\tilde{f}((z - z_0)/a)$. Six bounded derivatives of $\tilde{f}$ give $|\tilde{f}(\xi)| \leq C(1 + |\xi|^2)^{-3}$ by integration by parts. This bound is integrable in four dimensions, so $\|\tilde{f}\|_1 \leq C$, and hence $\|f\|_1 \leq CM$. The bounded-symbol Fourier formula gives the assertion; smoothing remainders are controlled by $\|f\|_1$. This estimate uses the derivative bounds (3.2), not general L^∞ boundedness.

Lemma 3.2. *Choose normal coordinates centered at x , on a uniformly small fixed ball, and put $t = d_g(x, y)$. Let a smooth radial cutoff be one on a smaller ball and vanish outside the coordinate ball. For $s > 0$ small, let u_s, h_s be this cutoff times*

$$\log \sqrt{s^2 + t^2}, \quad \sqrt{s^2 + t^2},$$

respectively, extended by zero. Near the center,

$$|BRdd_J^c u_s| \leq \frac{C}{s+t}, \quad |BRdd_J^c h_s| \leq C(1 + |\log(s+t)|). \quad (3.3)$$

Both expressions are uniformly bounded at any fixed positive distance from x .

Proof. The first-order property of Rdd_J^c gives

$$|\partial^j(Rdd_J^c u_s)| \leq C_j(s+t)^{-1-j}, \quad |\partial^j(Rdd_J^c h_s)| \leq C_j(s+t)^{-j}$$

near x . Away from x , all fixed-order derivatives are uniformly bounded, including terms from the cutoff.

At y , put $a = s + t$, assuming a small. Localize the input to a ball about y of radius a small fixed multiple of a . On this ball $s + d_g(x, \cdot)$ is comparable to a , so (3.2) gives contributions C/a and C . For the rest of the input within distance $O(a)$ of x , the kernel is bounded by Ca^{-4} . The respective input L^1 bounds on this region are Ca^3 and Ca^4 , again giving C/a and C .

On the more distant shells about x , with radius $\rho \geq Ca$, the kernel is $O(\rho^{-4})$, the input bounds are C/ρ and C , and the shell volume is $O(\rho^4)$. Summation over dyadic shells gives C/a and $C(1 + |\log a|)$. The logarithmic input has uniformly bounded total L^1 norm, since its local integral is at most $C \int_0^{\rho_0} \rho^3 / (s + \rho) d\rho \leq C\rho_0^3$. The square-root input has bounded L^1 norm as well. Thus the remaining smooth kernel terms are bounded uniformly. The same decomposition gives uniform bounds when y stays a fixed distance from the center. Compactness makes all chart and operator constants uniform in x . $\square$

Proof of Proposition 3.1. In the normal coordinates of Lemma 3.2, let $J_0 = J(x)$ be the constant complex structure, orthogonal for the Euclidean metric at the center. On the inner ball, the Hessians of the two radial functions are

$$\frac{I}{s^2 + t^2} - \frac{2zz^\top}{(s^2 + t^2)^2}, \quad \frac{I}{\sqrt{s^2 + t^2}} - \frac{zz^\top}{(s^2 + t^2)^{3/2}}.$$

Since $(z \cdot v)^2 + (z \cdot J_0 v)^2 \leq t^2 |v|^2$, their complex-line traces satisfy

$$dd_{J_0}^c u_s(v, J_0 v) \geq \frac{2s^2 |v|^2}{(s^2 + t^2)^2}, \quad (3.4)$$

$$dd_{J_0}^c h_s(v, J_0 v) \geq \frac{|v|^2}{\sqrt{s^2 + t^2}}. \quad (3.5)$$

The first lower bound is nonnegative everywhere and at least c/s^2 for $t < s$ and g -unit v .

Replacing J_0 by $J(y)$ both in the differential expression and in the second argument costs at most $C/(s+t)$ for u_s and C for h_s . To see this, use $J(y) - J_0 = O(t)$ on the Hessian terms and the bounded first derivatives of J on the gradient terms. Euclidean and g -norms are uniformly comparable.

Together with (3.3), this proves, for $s+t$ sufficiently small,

$$Du_s(v, Jv) \geq \frac{c}{s^2} \mathbf{1}_{\{t < s\}} - \frac{C}{s+t}, \quad Dh_s(v, Jv) \geq \frac{c'}{s+t}$$

with $c, c' > 0$. In the second estimate the logarithmic correction is absorbed by the positive inverse-distance term on a sufficiently small fixed neighborhood. Choose a fixed $A > 0$ large enough to absorb the negative term in the first estimate. Outside that neighborhood all terms are uniformly bounded. We obtain globally

$$D(u_s + Ah_s)(v, Jv) \geq \frac{c}{s^2} \mathbf{1}_{B_s(x)} - C'$$

for all unit v and small s .

The form on the left is closed and invariant. Applying P , which annihilates it, and using $\mu(X) = 1$, gives $0 \geq cs^{-2}\mu(B_s(x)) - C'$. This proves (3.1) for small s . Increasing the constant handles all larger radii. $\square$

4 Regularization and angular control

We now combine quadratic mass growth with regularized current pairings to prove that $Q \in L^2$ and to control the variation of nearby complex-line directions. These are the remaining quantitative inputs to the density-splitting theorem.

Let $\Delta = dd^* + d^*d$ be the Hodge Laplacian, and set

$$S_r = (1 + r^2 \Delta)^{-3}, \quad 0 < r < r_0.$$

The exponent is fixed. By Proposition 2.2, the regularizations of P, Q, T belong to L^2 : for fixed r , S_r has order -6 and maps H^{-3} into H^3 . The same is true for a positive current P' represented by any $\lambda' \leq \lambda$. Write μ' for the projection of λ' .

Lemma 4.1. *If T' is a closed current with $S_r T' \in L^2$, then*

$$\langle S_r T, *S_r T' \rangle_{L^2} = 0. \quad (4.1)$$

Proof. The operator S_r is self-adjoint, commutes with star in middle degree, and commutes across degrees with d and d^* . It preserves smooth forms. Thus $S_r T$ annihilates every smooth closed form by (2.4). Closedness of T' as a current means $d^* T' = 0$, so $*S_r T'$ is a distributionally closed L^2 form. Its heat regularizations are smooth closed forms and converge to it in L^2 . The asserted pairing follows by continuity. $\square$

4.1 The kernel and its positive leading part

For nearby x, y , let τ_{yx} denote metric parallel transport along the short geodesic from y to x .

Lemma 4.2. *The continuous kernel $K_r(x, y)$ of S_r^2 can be written*

$$K_r(x, y) = p_r(x, y)\tau_{yx} + E_r(x, y),$$

where $p_r \geq 0$ is supported in a fixed neighborhood of the diagonal. For every fixed positive integer N ,

$$0 \leq p_r(x, y) \leq C_N r^{-4} (1 + d_g(x, y)/r)^{-N}, \quad (4.2)$$

$$p_r(x, y) \geq c r^{-4} \quad (d_g(x, y) < r), \quad (4.3)$$

$$|E_r(x, y)| \leq C_N r^{-2} (1 + d_g(x, y)/r)^{-N}. \quad (4.4)$$

Proof. Spectral calculus gives

$$S_r^2 = \frac{1}{\Gamma(6)} \int_0^\infty e^{-s} s^5 e^{-r^2 s \Delta} ds. \quad (4.5)$$

The Hodge Laplacian is the Levi-Civita connection Laplacian on forms plus a zeroth-order curvature endomorphism. The leading term of its heat kernel, localized by a nonnegative diagonal cutoff, is

$$\chi(x, y) (4\pi t)^{-2} e^{-d_g(x, y)^2/(4t)} a(x, y) \tau_{yx},$$

where a is a smooth positive scalar volume correction and $\chi = 1$ sufficiently near the diagonal. For the leading coefficient, see the author version [15, Section 3, equation (3.4)]; for uniform short-time asymptotics and the global Gaussian bound, see [16, Theorems 1.1 and 3.5].

Fix a small $t_0 > 0$. After subtracting the displayed term, the short-time remainder has the bound

$$C t^{-1} e^{-c_0 d_g(x, y)^2/t} + C_M t^M, \quad 0 < t < t_0, \quad (4.6)$$

for any prescribed integer M , with constants depending on M . Indeed, take a sufficiently long heat parametrix: the further localized coefficients gain at least one factor of t , and its final uniform remainder can be made $O(t^M)$. Off the diagonal the cutoff errors are of arbitrarily high order.

Integrating the leading term for $r^2 s < t_0$ in (4.5) defines p_r . For each fixed N ,

$$e^{-c_0 d^2/(r^2 s)} \leq C_N (1 + \sqrt{s})^N (1 + d/r)^{-N}.$$

The resulting s -integrals converge: their powers at zero are s^3 for the leading term and s^4 for the Gaussian remainder, and e^{-s} controls infinity. This gives (4.2) and the Gaussian part of (4.4). Restricting to $1 \leq s \leq 2$ gives (4.3) for small r .

The smooth remainder in (4.6) integrates to $O(r^{2M})$. Since X has bounded diameter, choosing $2M \geq N - 2$ absorbs it into (4.4). For $s \geq t_0/r^2$, the heat kernel is uniformly bounded, and the tail of the gamma weight is exponentially small. This proves all the bounds. Notice that increasing N changes the parametrix length, not the fixed resolvent power. $\square$

For unit complex line bivectors at x, y , respectively ℓ, m , the algebra (2.1) gives the exact identity

$$\langle \ell, * \tau_{yx} m \rangle = \frac{1}{2} \langle F_x, \tau_{yx} F_y \rangle - \langle \ell, \tau_{yx} m \rangle = \frac{1}{2} |\ell - \tau_{yx} m|^2 - \frac{1}{4} |F_x - \tau_{yx} F_y|^2. \quad (4.7)$$

Here parallel transport is an isometry and commutes with star. In particular the negative term is $O(d_g(x, y)^2)$.

Lemma 4.3. *Uniformly for all $\lambda' \leq \lambda$,*

$$\|S_r P'\|_2 \leq C r^{-1}, \quad (4.8)$$

and, writing $d = d_g(x, y)$,

$$\begin{aligned} \langle S_r P, * S_r P' \rangle_{L^2} &\geq c r^{-4} \iint_{d < r} |\ell - \tau_{yx} m|^2 d\lambda(x, \ell) d\lambda'(y, m) \\ &\quad - C r^{-2} \iint (1 + d/r)^{-6} d\mu(x) d\mu'(y). \end{aligned} \quad (4.9)$$

Moreover

$$r^{-2} \int_X (1 + d_g(x, y)/r)^{-6} d\mu(x) \leq C. \quad (4.10)$$

Proof. The inner ball and the dyadic shells in Proposition 3.1 give (4.10): the shell at distance comparable to $2^j r$ contributes at most $C r^2 2^{-4j}$ before division by r^2 .

Self-adjointness and commutation with star give

$$\langle S_r P, * S_r P' \rangle = \iint \langle \ell, * K_r(x, y) m \rangle d\lambda(x, \ell) d\lambda'(y, m).$$

The identity is justified directly by the continuous kernel, or by approximation in H^{-3} : the operator S_r^2 has order -12 and maps H^{-3} into H^9 , so the dual pairings are continuous. Using (4.7), retain its nonnegative term only on $d < r$, where (4.3) applies. By (4.2) with $N \geq 8$, the negative term is bounded by $C r^{-2} (1 + d/r)^{-6}$; the kernel error has the same bound. This proves (4.9). For (4.8), use the analogous identity for $\|S_r P'\|_2^2$, take absolute values of the kernel, and apply (4.10) and $\mu' \leq \mu$. $\square$

The kernel estimate and quadratic growth bound the negative error in (4.9) uniformly. To extract an L^2 bound for Q , we must also control the cross terms between invariant and anti-invariant components. The next estimate supplies this control.

4.2 Mixing of invariant and anti-invariant components

Lemma 4.4 (Relative projection estimate). *Let Π be a fixed smooth orthogonal projection on $\Lambda^2 T^* X$. If a distributional two-form W satisfies $\Pi W = 0$ and $S_r W \in L^2$, then*

$$\|\Pi S_r W\|_2 \leq C r \|S_r W\|_2. \quad (4.11)$$

The constant may depend on Π , but is independent of r and W .

Proof. Put $U = S_r W$ and $A_r = (1 + r^2 \Delta)^3$. The operators S_r and A_r are inverse on distributions. Since $\Pi A_r U = \Pi W = 0$, we have, distributionally,

$$\Pi U = S_r A_r (\Pi U) = S_r [A_r, \Pi] U.$$

On smooth forms the adjoint of the operator on the right is $-[A_r, \Pi]S_r$. Expanding the commutator gives terms $\binom{3}{j}r^{2j}[\Delta^j, \Pi]S_r$, $1 \leq j \leq 3$. The principal symbol of Δ^j is scalar, so $[\Delta^j, \Pi]$ has differential order at most $2j - 1$. The elliptic Sobolev calculus [17, Propositions 6.7–6.8], together with spectral calculus for the Hodge Laplacian, gives

$$\|S_r\|_{L^2 \rightarrow H^{2j-1}} \leq Cr^{-(2j-1)}.$$

Each term of the adjoint therefore has norm $O(r)$. Taking its bounded Hilbert-space adjoint extends $S_r[A_r, \Pi]$ to L^2 , agreeing with its distributional action. Applying this bound to U proves (4.11). $\square$

Proposition 4.5. *The correction Q belongs to L^2 , and*

$$\mathcal{I}_r := \iint_{d_g(x,y) < r} |\ell - \tau_{yx}m|^2 d\lambda(x, \ell) d\lambda(y, m) \leq Cr^4. \quad (4.12)$$

For every positive current P' represented by $\lambda' \leq \lambda$,

$$\lim_{r \downarrow 0} \langle S_r P', *S_r Q \rangle_{L^2} = 0. \quad (4.13)$$

Proof. For any anti-invariant distribution V with $S_r V \in L^2$, split both factors along R and $1 - R$, and use Lemmas 4.3 and 4.4:

$$|\langle S_r P', S_r V \rangle| \leq Cr \|S_r P'\|_2 \|S_r V\|_2 \leq C \|S_r V\|_2. \quad (4.14)$$

Since Q is self-dual, Lemma 4.1 with $T' = T$ gives

$$0 = \langle S_r P, *S_r P \rangle + 2 \langle S_r P, S_r Q \rangle + \|S_r Q\|_2^2.$$

Equations (4.9), (4.10), and (4.14) imply

$$cr^{-4} \mathcal{I}_r + \|S_r Q\|_2^2 \leq C + C \|S_r Q\|_2.$$

Thus $\|S_r Q\|_2$ is uniformly bounded and $\mathcal{I}_r \leq Cr^4$. A weakly convergent L^2 subsequence, together with distributional convergence $S_r Q \rightarrow Q$, proves $Q \in L^2$.

For any anti-invariant $V \in L^2$, (4.14) and the L^2 contraction property of S_r give a uniform bound by $C\|V\|_2$. If V is smooth, then it is self-dual and

$$\langle S_r P', *S_r V \rangle = P'(S_r^2 V) \rightarrow P'(V) = 0,$$

because $S_r^2 V \rightarrow V$ smoothly. Smooth anti-invariant forms are dense in the anti-invariant L^2 subspace: smooth first and then apply R . Approximation of Q now proves (4.13). $\square$

5 Densities and planar tangent measures

We isolate the geometric statement that will be used to finish the proof. In its hypotheses, P need not annihilate closed invariant forms.

Theorem 5.1 (Closedness of the positive-density part). *Fix a smooth almost complex structure J and a J -Hermitian metric on a closed four-manifold. Suppose that*

$$P(\alpha) = \int_{\mathcal{C}} \alpha_x(\ell) d\lambda(x, \ell)$$

for a finite positive measure λ , with projection μ . Suppose also that $Q \in L^2$ is anti-invariant and that $P + Q$ is a closed current. Assume the estimates

$$\mu(B_r(x)) \leq Cr^2, \quad (5.1)$$

$$\iint_{d_g(x,y) < r} |\ell - \tau_{yx}m|^2 d\lambda(x, \ell) d\lambda(y, m) \leq Cr^4 \quad (5.2)$$

for all sufficiently small r , uniformly in x . Then the limit

$$\theta(x) = \lim_{r \downarrow 0} r^{-2} \mu(B_r(x))$$

exists at every point. If $E = \{x : \theta(x) > 0\}$, the current $P_E = \mathbf{1}_E P$ is closed.

For positive currents that are already closed, Elkhadhra proves that restriction to a J -analytic subset remains closed [4, Section 3.2, Lemma 1, author version]. Here only $P + Q$ is initially closed, and the positive-density set is not assumed to be J -analytic; the mass and angular estimates provide the cutoff control needed for the restriction.

Throughout this section and Section 6, assume the hypotheses of Theorem 5.1, and write $T = P + Q$. We continue to write $\mathcal{I}_r$ for the left side of (5.2). We first establish the density limit and show that, at μ -almost every positive-density point, the current has planar tangent measures. This makes the average transverse displacement small without supplying a rate. Section 6 combines that smallness with the angular bound to prove closedness.

Lemma 5.2. *The two-density $\theta(p)$ exists and is finite at every $p \in X$.*

The radial comparison is in the tradition of Lelong–Jensen formulas for almost complex currents; see [5, Proposition 1 and Corollary 2]. We apply closure to $P + Q$ and estimate the L^2 correction directly.

Proof. The current T has measure coefficients with variation bounded by a constant times $\sigma = \mu + |Q| dV_g$. Cauchy–Schwarz and (5.1) give

$$\sigma(B_s(p)) \leq Cs^2,$$

uniformly in p .

In normal coordinates z at p , let $J_0 = J(p)$ and $F_0 = F_p$ be constant, and put $t = |z| = d_g(p, y)$. Our sign convention gives $dd_{J_0}^c(t^2/2) = 2F_0$. Call a radius *good* if its sphere is σ -null. For good radii define

$$b(r) = r^{-2} T(\mathbf{1}_{B_r(p)} F_0).$$

These pairings are well-defined using the coefficient measures. We claim that

$$b(s) - b(r) = \frac{1}{2} T(\mathbf{1}_{\{r < t < s\}} dd_{J_0}^c \log t), \quad 0 < r < s. \quad (5.3)$$

To verify it, define

$$\phi_a(t) = \begin{cases} \log a + (t^2 - a^2)/(2a^2), & t < a, \\ \log t, & t \geq a. \end{cases}$$

The function $\phi_r - \phi_s$ is compactly supported and $C^{1,1}$. Its $dd_{J_0}^c$ is $2(r^{-2} - s^{-2})F_0$ on B_r , equals $dd_{J_0}^c \log t - 2s^{-2}F_0$ on the annulus, and is zero outside B_s . For fixed r, s , compactly supported smooth approximations have uniformly bounded Hessians and converge in second derivatives off

the two spheres. Dominated convergence against σ permits testing closure on their exact two-forms and passing to the limit. The resulting identity is (5.3).

The form $dd_{J_0}^c \log t$ is J_0 -invariant and semipositive, with size $O(t^{-2})$. Since $J(y) - J_0 = O(t)$, it evaluates on actual complex line bivectors to at least $-C/t$, and its J -anti-invariant projection has size $O(1/t)$. The latter estimate controls its pairing with Q . Inward dyadic shells and the quadratic bound on σ give

$$\int_{\{0 < t < s\}} t^{-1} d\sigma \leq Cs.$$

It follows that

$$b(s) - b(r) \geq -Cs. \quad (5.4)$$

Also $T(\mathbf{1}_{B_r} F) = \mu(B_r)$, since $\langle Q, F \rangle_g = 0$ almost everywhere. The bound $F - F_0 = O(t)$ therefore implies

$$|b(r) - r^{-2} \mu(B_r(p))| \leq Cr. \quad (5.5)$$

In particular b is bounded. Fix a good radius s in (5.4) and let $r \downarrow 0$ through good radii; then let $s \downarrow 0$ along a sequence realizing the lower limit of b . The upper and lower limits agree. Equation (5.5) gives the density limit along good radii.

Only countably many spheres at a fixed center can have positive σ -mass. Any radius r can therefore be bracketed by good radii in $((1 - \varepsilon)r, r)$ and $(r, (1 + \varepsilon)r)$. Ball inclusion, followed by $r \downarrow 0$ and then $\varepsilon \downarrow 0$, proves the asserted limit along all radii. Its finiteness follows from (5.1). $\square$

For $d_g(x, y)$ below the injectivity radius, set $z_x(y) = \exp_x^{-1}(y)$.

Lemma 5.3 (Vanishing transverse moment). *Under the hypotheses of Theorem 5.1,*

$$\mathcal{N}_r := \int_{\mathcal{C}} \int_{B_r(x)} \left| \text{pr}_{L_t^\perp} \frac{z_x(y)}{r} \right|^2 d\mu(y) d\lambda(x, \ell) = o(r^2). \quad (5.6)$$

Proof. Write $P = A\mu$ in a smooth local frame, so that $A(p)$ is the conditional average of the line bivectors over p . At μ -almost every point, differentiation with respect to the Radon measure μ gives

$$r^{-2} \int_{B_{Dr}(p)} |A(y) - A(p)| d\mu(y) \longrightarrow 0 \quad (D < \infty \text{ fixed}). \quad (5.7)$$

For the precise differentiation input one may use [8, Theorem 4.33 and Corollary 4.34]. To see the normalization, enclose $B_{Dr}(p)$ in a Euclidean coordinate ball of radius CDr . The mean oscillation on that ball tends to zero, while its mass divided by r^2 is bounded by (5.1). This does not require μ to be doubling. Smooth changes of frame add a vanishing $O(r)$ error after this normalization.

Fix such a differentiation point p with $\theta(p) > 0$. The locally rescaled measures

$$\nu_r = r^{-2} (z_p/r)_* \mu$$

have uniformly bounded mass on each compact subset of $T_p X$, so every sequence of scales has a vaguely convergent subsequence. Every limit ν satisfies

$$\nu(B_D(0)) = \theta(p) D^2 \quad (D > 0). \quad (5.8)$$

Initially the identity follows at continuity radii of ν . Increasing continuity radii recover the mass of each open ball, while decreasing continuity radii recover the corresponding closed ball. Both masses are $\theta(p) D^2$, so the identity holds for every $D > 0$ and every centered sphere is ν -null.

The current and measure rescalings have the same normalization. Indeed, put $D_r(y) = z_p(y)/r$. For a compactly supported test two-form $\beta = \sum_{i<j} \beta_{ij}(w) dw_i \wedge dw_j$ on $T_p X$,

$$D_r^* \beta = r^{-2} \sum_{i<j} \beta_{ij}(z_p/r) dz_i \wedge dz_j.$$

Thus the pushforward $(D_r)_* T$, defined by testing T against $D_r^* \beta$, already has the r^{-2} factor in its coefficient measures; no further normalization is used. Equation (5.7) makes its P part converge to $A(p)v$. Its Q part vanishes in variation on each fixed ball, since

$$r^{-2} \int_{B_{Dr}(p)} |Q| dV_g \leq C_D \|Q\|_{L^2(B_{Dr}(p))} \longrightarrow 0. \quad (5.9)$$

The last limit holds at every point by absolute continuity of the integral of $|Q|^2$. Compactly supported tests in the rescaled coordinates pull back inside the original coordinate chart for small r . Thus closure passes to the limit, and $A(p)v$ is a closed constant-polarization current on $T_p X$.

For a constant skew matrix A , this closure says $\sum_i A^{ij} \partial_i v = 0$ for each j . Consequently $\partial_v v = 0$ for every $v \in \text{im } A(p)$. For a compactly supported smooth function ϕ , the derivative of $t \mapsto \int \phi(w + tv) dv(w)$ is therefore zero. This proves that v is invariant under translations along $\text{im } A(p)$. The normalization $F_p(A(p)) = 1$ excludes rank zero. Rank four would imply that v is a constant multiple of four-dimensional Lebesgue measure, contrary to (5.8). Therefore $A(p)$ has real rank two.

The associated positive Hermitian matrix has trace one and complex rank one. Thus $A(p)$ is itself a unit complex line bivector. If λ_p is the conditional probability measure on $\mathcal{C}_p$, then

$$\int_{\mathcal{C}_p} |\ell - A(p)|^2 d\lambda_p(\ell) = 1 - |A(p)|^2 = 0.$$

Hence λ_p is concentrated on $A(p)$. Denote its plane by L .

Translation invariance along L implies

$$v = \mathcal{L}_L^2 \otimes \gamma$$

for a positive Radon measure γ on $L^\perp$. For example, testing against a nonnegative compactly supported function in the normal variables gives a translation-invariant measure in the tangential variables, hence a multiple of area; these multiples define γ . Dividing the centered ball mass by D^2 in (5.8) yields

$$\theta(p) = \pi \int_{L^\perp} (1 - |w|^2/D^2)_+ d\gamma(w). \quad (5.10)$$

For $D_2 > D_1$, the difference of these integrands is nonnegative and strictly positive at every $0 < |w| < D_2$. Since the integrals are equal, γ is supported at the origin. Every tangent limit is therefore $(\theta(p)/\pi) \mathcal{L}_L^2$.

It follows that the normalized inner integral in (5.6) tends to zero at λ -almost every (p, ℓ) of positive density. Indeed, in the rescaled coordinates a continuous cutoff equal to one on B_1 , supported in B_2 , times $|\text{pr}_{L^\perp} z|^2$ dominates the integrand and vanishes on every tangent limit. Subsequential compactness gives convergence along all scales.

When $\theta(p) = 0$, the same conclusion follows directly by bounding the normalized inner integral by $r^{-2} \mu(B_r(p))$. These normalized integrals are uniformly bounded by (5.1). Dominated convergence against the finite measure λ proves (5.6). $\square$

6 Closedness of the density restriction

Proof of Theorem 5.1. The existence of the density is Lemma 5.2. We approximate the positive-density restriction by smooth functions of a regularized density. Closedness of T then reduces the problem to controlling full derivatives against Q and tangential derivatives against P . The tangential estimate combines the quantitative angular bound with the vanishing transverse moment from Lemma 5.3; their product will make the cutoff boundary tend to zero. Choose a nonnegative, smooth, nonincreasing function $h : [0, \infty) \rightarrow [0, \infty)$, positive near zero and zero on a neighborhood of $[1, \infty)$. For r below the injectivity radius define

$$f_r(x) = r^{-2} \int_X h(d_g(x, y)^2 / r^2) d\mu(y). \quad (6.1)$$

The kernel is smooth near the diagonal and vanishes before the cut locus, so f_r is smooth. It is uniformly bounded. Writing $H(t) = h(t^2)$, integration of ball indicators gives

$$f_r(x) = \int_0^1 [-H'(t)] r^{-2} \mu(B_{rt}(x)) dt \longrightarrow c_h \theta(x), \quad c_h = \int_0^1 h(u) du > 0.$$

The quadratic growth bound supplies domination for this limit. In particular θ is Borel measurable. Differentiation of the finite measure μ with respect to four-dimensional volume also shows that

$$\theta(x) = 0 \quad \text{for volume-almost every } x. \quad (6.2)$$

Indeed, apply the differentiation theorem for Radon measures [8, Theorem 4.19] in a fixed coordinate chart, with smooth volume as the denominator measure. At volume-almost every center the measure-to-volume ratios of sufficiently small coordinate balls are bounded. Enclosing a geodesic ball of radius r in a coordinate ball of comparable radius therefore gives $\mu(B_r(x)) = O_x(r^4)$, and hence $r^{-2} \mu(B_r(x)) \rightarrow 0$.

The error from the L^2 correction

Put $M_r(x) = \mu(B_r(x))$. Cauchy–Schwarz gives

$$r^{-3} \int_X |Q| M_r dV_g \leq \left(\int_X |Q|^2 r^{-2} M_r dV_g \right)^{1/2} \left(\int_X r^{-4} M_r dV_g \right)^{1/2} \longrightarrow 0. \quad (6.3)$$

For the second factor, Fubini and the four-dimensional volume bound give

$$\int_X r^{-4} M_r(x) dV_g(x) = r^{-4} \int_X \text{vol}_g(B_r(y)) d\mu(y) \leq C\mu(X).$$

For the first, $r^{-2} M_r$ is bounded and tends to zero volume-almost everywhere by (6.2). Dominated convergence applies to $|Q|^2$.

First variation gives $|df_r(x)| \leq Cr^{-3} M_r(x)$. Thus (6.3) also proves

$$\int_X |Q| |df_r| dV_g \longrightarrow 0. \quad (6.4)$$

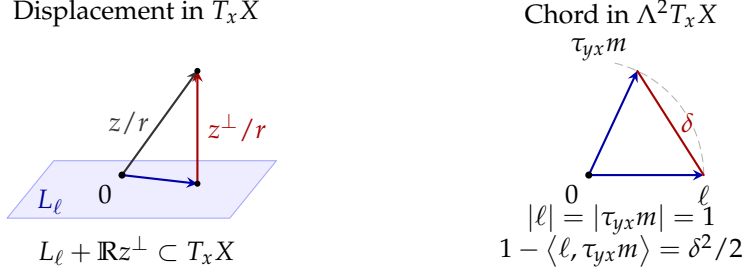


Figure 1: The two factors in the tangential cutoff estimate. The left panel lies in the subspace spanned by L_ℓ and the normal displacement. The right panel lies in the span of two unit bivectors, in a different ambient vector space. Closedness produces the product $\delta|z^\perp|/r$. The squared integrals of its factors are $\mathcal{I}_r = O(r^4)$ and $\mathcal{N}_r = o(r^2)$, respectively.

Tangential derivatives of the density cutoff

We prove the complementary estimate

$$\int_{\mathcal{C}} |df_r(x)|_{L_\ell} d\lambda(x, \ell) \longrightarrow 0, \quad (6.5)$$

where the subscript means restriction of the covector to the two-plane L_ℓ .

For $(x, \ell) \in \mathcal{C}$, $y \in B_r(x)$, and $m \in \mathcal{C}_y$, put

$$z = z_x(y), \quad z^\perp = \text{pr}_{L_\ell^\perp} z, \quad \delta = |\ell - \tau_{yx} m|.$$

These quantities measure different errors (Figure 1). Closedness will replace a derivative along L_ℓ by normal derivatives. Differentiating the radial kernel normally contributes $|z^\perp|/r$, while the resulting mixed two-forms contribute δ . Keeping these factors together is what makes their integrated product overcome the cutoff's r^{-3} normalization.

Fix (x, ℓ) . Choose orthonormal normal coordinates $z = z_x(y)$ with $\ell = e_1 \wedge e_2$, and put $h_r(z) = h(|z|^2/r^2)$. At the fixed center x , first variation of squared distance gives $d_x[d_g(x, y)^2](e_i) = -2z_i$. Hence, for $i = 1, 2$,

$$df_r(x)e_i = -r^{-2} \int_X \partial_{z_i} h_r d\mu(y). \quad (6.6)$$

This differentiates the scalar distance kernel. We now freeze x, ℓ , and these coordinates, and use closedness of T in the y variable.

Coordinate covectors in normal coordinates differ from radially parallel covectors by $O(|z|^2)$. Indeed, the Jacobi equation along $t \mapsto \exp_x(tz)$, in a parallel frame, has curvature coefficient $O(|z|^2)$. The field with initial data $(0, v)$ is therefore $tv + O(|z|^2|v|)$ for $0 \leq t \leq 1$, giving $\tau_{yx} \circ d(\exp_x)_z = \text{id} + O(|z|^2)$. Inverting gives the same estimate for coordinate covectors. Since both line bivectors have norm one,

$$(dz_1 \wedge dz_2)(m) = \langle \ell, \tau_{yx} m \rangle + O(r^2) = 1 - \frac{1}{2}\delta^2 + O(r^2) \quad (|z| < r). \quad (6.7)$$

All errors are uniform. Comparing (6.6) with the P part of the current below, and bounding its Q part, gives

$$\begin{aligned} |df_r(x)e_i + r^{-2}T((\partial_{z_i} h_r) dz_1 \wedge dz_2)| &\leq Cr^{-3} \int_{d_g(x, y) < r} (\delta^2 + r^2) d\lambda(y, m) \\ &\quad + Cr^{-3} \int_{B_r(x)} |Q| dV_g. \end{aligned} \quad (6.8)$$

After integration against $\lambda(x, \ell)$, its first term is $O(r)$ by (5.2). Its r^2 term is $O(r)$ by (5.1). Its last term tends to zero by Fubini and (6.3). Thus the integrated comparison error vanishes.

For $i = 1$, apply closure to $d(h_r dz_2)$; for $i = 2$, apply it to $d(h_r dz_1)$. These one-forms are compactly supported in the chart and extend smoothly by zero. Since $d(dz_j) = 0$, each $T((\partial_{z_i} h_r) dz_1 \wedge dz_2)$ is, up to sign, a sum of terms

$$T((\partial_{z_a} h_r) dz_a \wedge dz_j), \quad a = 3, 4, \quad j \in \{1, 2\}.$$

For the normal derivatives,

$$|\partial_{z_a} h_r| \leq Cr^{-1} \left| \text{pr}_{L_\ell^\perp}(z/r) \right|.$$

The mixed coordinate two-forms satisfy

$$|(dz_a \wedge dz_j)(m)| \leq C(\delta + r^2).$$

Indeed their radially parallel counterparts vanish on ℓ , and their coordinate error is $O(r^2)$.

The integrated Q and r^2 errors are controlled as in (6.8). Including the outer factor r^{-2} , the remaining term is at most

$$Cr^{-3} \iint_{d_g(x, y) < r} \delta \left| \text{pr}_{L_\ell^\perp}(z_x(y)/r) \right| d\lambda(y, m) d\lambda(x, \ell).$$

By Cauchy–Schwarz, (5.2), and Lemma 5.3, this is bounded by

$$Cr^{-3} \sqrt{\mathcal{I}_r} \sqrt{\mathcal{N}_r} = r^{-3} O(r^2) o(r) = o(1). \quad (6.9)$$

The second squared integral is exactly $\mathcal{N}_r$, since its integrand does not depend on m . Combining with (6.8) proves (6.5). No derivative of a center-dependent frame occurs: first variation was applied before the fixed-chart current tests.

Passing to the positive-density set

Let $b : \mathbb{R} \rightarrow \mathbb{R}$ be smooth with $b(0) = 0$. On the uniformly bounded range of f_r , both b and b' are bounded. Dominated convergence gives

$$b(f_r)T \longrightarrow b(c_h \theta)P \quad (6.10)$$

as currents. For P this follows from pointwise convergence of f_r . For Q it follows from (6.2) and $Q \in L^1$.

If η is a smooth one-form, closedness of T and the product rule give

$$(b(f_r)T)(d\eta) = -T(b'(f_r) df_r \wedge \eta).$$

The Q contribution tends to zero by (6.4). A line bivector ℓ evaluates $df_r \wedge \eta$ using only the restriction of df_r to L_ℓ , so the P contribution tends to zero by (6.5). It follows from (6.10) that $b(c_h \theta)P$ is closed.

Finally take $b_n(s) = 1 - e^{-ns}$. On the relevant nonnegative range, $0 \leq b_n \leq 1$ and $b_n(c_h \theta) \rightarrow \mathbf{1}_E$. A second dominated-convergence limit proves that $P_E = \mathbf{1}_E P$ is closed. The limits are successive: first $r \downarrow 0$ for each fixed n , then $n \rightarrow \infty$. No bound on b'_n uniform in n is required. $\square$

7 The final pairing

Proof of Theorem 1.1. Let ω be a smooth closed form taming J . Suppose for contradiction that there is no compatible symplectic form, and take P, Q, T, λ, μ from Proposition 2.2. Propositions 3.1 and 4.5 verify all the hypotheses of Theorem 5.1. Consequently the current $P_E = \mathbf{1}_{\{\theta > 0\}} P$ is closed. Set

$$P' = \mathbf{1}_{X \setminus E} P, \quad T' = P' + Q = T - P_E.$$

Then T' is closed. The current P' is represented by the restriction $\lambda' \leq \lambda$ to base points outside E , with projection $\mu' = \mathbf{1}_{X \setminus E} \mu$. We pair T with this residual current rather than with itself: placing the second positive factor on zero-density centers makes the previously bounded negative error in the pairing tend to zero.

We first show that the negative term in (4.9) tends to zero for this choice of P' . For each $y \notin E$,

$$r^{-2} \int_X (1 + d_g(x, y)/r)^{-6} d\mu(x) \longrightarrow 0. \quad (7.1)$$

On the inner ball and on every fixed dyadic shell this follows from $s^{-2} \mu(B_s(y)) \rightarrow 0$. The quadratic mass bound dominates the shell contributions by $C2^{-4j}$, a summable sequence, giving the full limit. The uniform bound (4.10) then permits dominated convergence against μ' . Discarding the nonnegative angular term in (4.9) yields

$$\liminf_{r \downarrow 0} \langle S_r P, *S_r P' \rangle_{L^2} \geq 0. \quad (7.2)$$

Apply Lemma 4.1 to T' . Its regularizations are in L^2 , since P' has measure coefficients and $Q \in L^2$. Expanding gives

$$0 = \langle S_r P, *S_r P' \rangle + \langle S_r P, *S_r Q \rangle + \langle S_r Q, *S_r P' \rangle + \|S_r Q\|_2^2.$$

Both mixed terms tend to zero by (4.13), applied once to P and once to P' . Since $S_r Q \rightarrow Q$ strongly in L^2 , (7.2) implies $0 \geq \|Q\|_2^2$. Therefore $Q = 0$.

It follows that $P = T$ annihilates all smooth closed forms. In particular $P(\omega) = 0$. On the other hand, $\omega_x(\ell) > 0$ on the compact bundle $\mathcal{C}$; it has a positive minimum there. Equation (2.2) and $\lambda(\mathcal{C}) = 1$ give $P(\omega) > 0$, a contradiction.

The positive cone and the subspace of closed invariant smooth forms therefore intersect. Any form in their intersection is the required smooth compatible symplectic form for J . $\square$

8 The compatible cone

The distinction between existence and realization of a specified class appears in the separating functional. For existence, it annihilates all closed invariant forms. For a specified class, it need only annihilate exact invariant forms. We first establish the estimates in this setting, then identify the classes of the resulting currents.

8.1 Invariant representatives and separation

We retain the Hermitian metric, the projection R , and the closed lift $B = dd^*G + H$ of Lemma 2.1. For a closed current D , let $\mathfrak{c}(D) \in H^2(X; \mathbb{R})$ denote its intersection-dual class:

$$D(\alpha) = \mathfrak{c}(D) \cdot [\alpha] \quad \text{for every smooth closed two-form } \alpha.$$

Equivalently, $\mathfrak{c}(D)$ is represented by the Hodge star of the harmonic part of the distributional two-form associated with D .

Lemma 8.1. *The classes represented by closed invariant forms are exactly $\mathcal{V} = (H_J^-)^\perp$. The intersection form is positive definite on H_J^- , and*

$$H^2(X; \mathbb{R}) = \mathcal{V} \oplus H_J^-.$$

The operator $1 - BR$ takes exact forms to exact invariant forms.

Proof. Closed anti-invariant forms are self-dual, hence harmonic. Their intersection pairing is their L^2 inner product, so it is positive definite on their classes. Invariant and anti-invariant forms wedge to zero pointwise.

For any closed two-form α , the form $(1 - BR)\alpha$ is closed and invariant, and

$$[\alpha] = [(1 - BR)\alpha] + [HR\alpha],$$

because $BR\alpha = dd^*G\alpha + HR\alpha$. The second summand is represented by a closed anti-invariant form. The two subspaces are orthogonal, and positive definiteness on the second makes their intersection zero. This proves the decomposition and the assertion about $\mathcal{V}$.

If α is exact, it is L^2 -orthogonal to every harmonic form. For a harmonic anti-invariant form β , $\langle R\alpha, \beta \rangle_{L^2} = \langle \alpha, \beta \rangle_{L^2} = 0$. Thus $HR\alpha = 0$, and both $BR\alpha$ and $(1 - BR)\alpha$ are exact. $\square$

A current is called *positive* here if it annihilates anti-invariant forms and is nonnegative on pointwise semipositive invariant forms. As in Proposition 2.2, it has a representation by a finite positive measure on $\mathcal{C}$. Its trace measure is the projection to X .

Lemma 8.2 (Prescribed-class separation). *Let $a \in \mathcal{V}$ have no compatible representative. There are currents $P, Q, T \in H^{-3}$, with P nonzero and positive, such that*

$$P(F) = 1, \quad P \text{ annihilates exact invariant forms}, \quad (8.1)$$

$$T = P + Q \text{ is closed}, \quad RQ = Q = *Q, \quad (8.2)$$

$$\mathfrak{c}(T) \in \mathcal{V}, \quad \mathfrak{c}(T) \cdot a \leq 0. \quad (8.3)$$

Moreover, a class $b \in H^2(X; \mathbb{R})$ is taming if and only if

$$b \cdot \mathfrak{c}(N) > 0 \quad \text{for every nonzero closed positive current } N.$$

Proof. By Lemma 8.1, the closed invariant representatives of a form a nonempty affine space. Separate this affine space from the open cone of positive invariant forms in the Fréchet space of smooth invariant forms. Hahn–Banach separation gives a nonzero continuous functional, nonnegative on the positive cone and nonpositive on the affine space. It vanishes on the direction space, which consists of exact invariant forms. Indeed, translating by any real multiple of a direction preserves the affine space, while positive scalings preserve the cone. Extend the functional by $1 - R$ and normalize it as in Proposition 2.2 to obtain $P(F) = 1$. The same positivity argument gives its measure representation.

Define $T(\alpha) = P((1 - BR)\alpha)$ and $Q(\alpha) = -P(BR\alpha)$. Lemma 8.1 shows that T annihilates every exact form, so it is closed. The projection identities give $RQ = Q = *Q$. A closed anti-invariant form β lies in the kernel of the operator used to construct G ; hence $B\beta = \beta$ and $T(\beta) = 0$. Thus $\mathfrak{c}(T) \in \mathcal{V}$. On a closed invariant representative of a , the currents T and P agree, proving (8.3). The Sobolev assertion follows from the measure representation and the order-zero lift, as in Proposition 2.2.

For the last statement, a taming representative evaluates strictly positively on every nonzero positive current. Conversely, if b has no taming representative, separate its affine space of closed representatives from the open cone of forms positive on complex lines. The separator annihilates all exact forms. Since arbitrary anti-invariant forms can be added to that cone, it also annihilates all anti-invariant forms. It is therefore a nonzero closed positive current N , and $b \cdot \mathfrak{c}(N) \leq 0$. $\square$

8.2 Pairings for currents of arbitrary class

Lemma 8.3. *For closed currents $D_1, D_2 \in H^{-3}$,*

$$\langle S_r D_1, *S_r D_2 \rangle_{L^2} = \mathfrak{c}(D_1) \cdot \mathfrak{c}(D_2). \quad (8.4)$$

Proof. The forms $*S_r D_i$ lie in H^3 , are distributionally closed, and have the harmonic components representing $\mathfrak{c}(D_i)$. Here we use commutation with d^* , $*$, and the fact that S_r is the identity on harmonic forms. Their heat regularizations are smooth closed representatives of these classes. The intersection identity for smooth forms passes to the limit in L^2 . In degree two on a four-manifold, $\alpha \wedge \beta = (*\alpha) \wedge (*\beta)$, giving (8.4). $\square$

We record the estimates that apply to the separator in Lemma 8.2. In the next proposition, positive currents are not required to have any prescribed cohomology class.

Proposition 8.4. *Suppose that $P \neq 0$ is positive, annihilates exact invariant forms, and $T = P + Q$ is closed, where $Q \in H^{-3}$ is anti-invariant. Write λ for a positive measure representing P , and μ for its trace. Then*

$$\mu(B_s(x)) \leq Cs^2, \quad Q \in L^2, \quad (8.5)$$

$$\iint_{d_g(x,y) < r} |\ell - \tau_{yx} m|^2 d\lambda(x, \ell) d\lambda(y, m) \leq Cr^4. \quad (8.6)$$

Consequently the positive-density restriction P_E is closed, where $E = \{x : \lim_{s \downarrow 0} s^{-2} \mu(B_s(x)) > 0\}$.

If A is any positive current whose trace has quadratic growth, then

$$\lim_{r \downarrow 0} \langle S_r A, *S_r Q \rangle_{L^2} = 0. \quad (8.7)$$

Every closed positive current has quadratic trace growth.

Proof. Each radial test $Du = (1 - BR)dd^c \mu$ in Section 3 is exact by Lemma 8.1. Its pointwise estimates therefore give the quadratic growth bound under the present annihilation hypothesis. The normalization $P(F) = 1$ is inessential: for arbitrary finite mass it only changes the constant. In particular, this argument applies to every closed positive current.

We will use the kernel estimates for any pair A_1, A_2 of positive currents with quadratic-growth trace measures μ_1, μ_2 . Write λ_1, λ_2 for their representing measures. The proof of Lemma 4.3, using Lemma 4.2 and (4.7), gives

$$\|S_r A_i\|_2 \leq C_i r^{-1}, \quad (8.8)$$

$$\begin{aligned} \langle S_r A_1, *S_r A_2 \rangle_{L^2} &\geq cr^{-4} \iint_{d < r} |\ell - \tau_{yx} m|^2 d\lambda_1(x, \ell) d\lambda_2(y, m) \\ &\quad - Cr^{-2} \iint (1 + d/r)^{-6} d\mu_1(x) d\mu_2(y), \end{aligned} \quad (8.9)$$

where $d = d_g(x, y)$. These statements do not require either measure to be dominated by the other. The shell bound (4.10) holds separately for each μ_i , proving the norm bound and a uniform bound on the negative term.

For any anti-invariant distribution W with $S_r W \in L^2$, Lemma 4.4 and (8.8) give

$$|\langle S_r A_i, *S_r W \rangle_{L^2}| \leq Cr \|S_r A_i\|_2 \|S_r W\|_2 \leq C'_i \|S_r W\|_2. \quad (8.10)$$

Apply (8.4) to T, T . Since Q is self-dual, expansion and these estimates yield

$$cr^{-4}\mathcal{I}_r + \|S_r Q\|_2^2 \leq C + |\mathfrak{c}(T)^2| + C\|S_r Q\|_2,$$

where $\mathcal{I}_r$ is the angular integral in (8.6). Hence $S_r Q$ is uniformly bounded in L^2 , and the angular bound follows. Weak compactness and distributional convergence identify an L^2 limit with Q .

For smooth anti-invariant W ,

$$\langle S_r A, *S_r W \rangle = A(S_r^2 W) \longrightarrow A(W) = 0.$$

The uniform bound (8.10), the L^2 contraction property of S_r , and approximation by smooth anti-invariant forms give (8.7). Finally, (8.5)–(8.6) verify the hypotheses of Theorem 5.1, which proves that P_E is closed. $\square$

The next lemma permits pairing the zero-density part with positive currents unrelated to the original separator.

Lemma 8.5 (Residual pairings). *Let A, P' be positive currents whose trace measures ν, μ' have quadratic growth. Suppose*

$$\lim_{s \downarrow 0} s^{-2} \mu'(B_s(x)) = 0 \quad \text{for } \mu'\text{-almost every } x.$$

Then

$$\liminf_{r \downarrow 0} \langle S_r A, *S_r P' \rangle_{L^2} \geq 0. \quad (8.11)$$

Proof. We first show

$$(\mu' \times \nu)\{(x, y) : d_g(x, y) < s\} = o(s^2). \quad (8.12)$$

For small s , choose an s -separated maximal set of centers and let B_j be their radius- s balls. They cover X , and their fixed dilates have uniformly bounded overlap, by the volume comparison in smooth coordinate charts. If $x \in B_j$ and $d_g(x, y) < s$, then $y \in 2B_j$. Thus

$$(\mu' \times \nu)\{d < s\} \leq \sum_j \mu'(B_j) \nu(2B_j).$$

Bounded overlap gives

$$\sum_j \mu'(B_j)^2 \leq C \int_X \mu'(B_{2s}(x)) d\mu'(x) = o(s^2).$$

Indeed, the integrand divided by s^2 is bounded by quadratic growth and tends to zero at μ' -almost every center. Also

$$\sum_j \nu(2B_j)^2 \leq Cs^2 \sum_j \nu(2B_j) = O(s^2).$$

Cauchy–Schwarz proves (8.12).

In (8.9), discard the angular term. The remaining negative error tends to zero: on $d < r$, and on each fixed shell $2^j r \leq d < 2^{j+1} r$, this follows from (8.12). Quadratic growth bounds the shell contributions by $C2^{-4j}$, allowing summation by dominated convergence. This proves (8.11). $\square$

8.3 Proof of the cone description

Proof of Theorem 1.2. Adding a closed anti-invariant form does not change positivity on complex lines. Lemma 8.1 therefore gives (1.1). In particular $\mathcal{C}$ is nonempty. It is a convex cone of positive-square classes in $\mathcal{V}$, since its classes have closed taming representatives.

A compatible form wedges strictly positively with every taming form. Indeed, in a unitary frame diagonalizing the compatible form, the wedge product is a positive linear combination of the two positive diagonal entries of the taming form's invariant part. Consequently every $a \in \mathcal{K}_j^c$ lies in $\mathcal{C}$ and pairs positively with every class in $\mathcal{C}$, hence nonnegatively with $\overline{\mathcal{C}}$.

The compatible cone is open in $\mathcal{V}$: add sufficiently small linear combinations of closed invariant representatives of a basis of $\mathcal{V}$ to a compatible form. The intersection form on $\mathcal{V}$ is nondegenerate by Lemma 8.1. If $a \cdot y = 0$ for a nonzero $y \in \overline{\mathcal{C}}$, a sufficiently small perturbation of a within the compatible cone would pair negatively with y . Thus every compatible class satisfies the strict positivity in (1.2).

Conversely, suppose $a \in \mathcal{C}$ has this strict positivity but no compatible representative. Apply Lemma 8.2 and Proposition 8.4. Let

$$P' = P - P_E, \quad T' = P' + Q = T - P_E, \quad y = c(T').$$

Both P_E and T' are closed. Since P_E is invariant, $c(P_E) \in \mathcal{V}$, and hence $y \in \mathcal{V}$. The trace μ' of P' satisfies the hypotheses of Lemma 8.5: it is bounded by μ and is concentrated where the latter has zero two-density.

We claim that $y \in \overline{\mathcal{C}}$. Let N be any nonzero closed positive current. Its trace has quadratic growth by Proposition 8.4. By (8.4),

$$y \cdot c(N) = \langle S_r N, *S_r P' \rangle + \langle S_r N, *S_r Q \rangle.$$

The mixed term tends to zero by (8.7), and Lemma 8.5 bounds the lower limit of the first term by zero. Thus $y \cdot c(N) \geq 0$. Since a is taming, $(y + \varepsilon a) \cdot c(N) > 0$ for every $\varepsilon > 0$. The last part of Lemma 8.2 shows that $y + \varepsilon a$ is taming. It also lies in $\mathcal{V}$, so $y + \varepsilon a \in \mathcal{C}$, proving the claim.

It follows that $a \cdot y \geq 0$, with strict inequality if $y \neq 0$. If $P' \neq 0$, then $y \neq 0$. To see this, suppose $y = 0$ and pair T' with itself using (8.4). We obtain

$$0 = \langle S_r P', *S_r P' \rangle + 2 \langle S_r P', *S_r Q \rangle + \|S_r Q\|_2^2.$$

The residual lemma, the mixed-term limit, and strong L^2 convergence of $S_r Q$ force $Q = 0$. Then the nonzero positive current $P' = T'$ evaluates strictly positively on a closed tamer, contrary to $c(T') = 0$.

Finally, a taming representative of a gives $a \cdot c(P_E) \geq 0$, strictly if $P_E \neq 0$. At least one of P_E, P' is nonzero. Therefore

$$a \cdot c(T) = a \cdot c(P_E) + a \cdot y > 0,$$

contradicting (8.3). This proves (1.2). $\square$

Proof of Corollary 1.3. The subspace H_j^- is positive definite, while its orthogonal complement $\mathcal{V}$ contains the nonempty positive-square cone $\mathcal{C}$. Hence $h_j^- \leq b_2^+ - 1$. If equality holds, the intersection form on $\mathcal{V}$ has signature $(1, b_2^-)$. Convexity puts $\mathcal{C}$ in one component of its positive-square cone. Every element of that component pairs strictly positively with every nonzero element of its closure, by Cauchy–Schwarz in coordinates of signature $(+, -, \dots, -)$. Thus all $a \in \mathcal{C}$ satisfy (1.2), and $\mathcal{C} = \mathcal{K}_j^c$. Equation (1.1) gives the asserted identity. If $b_2^+ = 1$, the inequality forces $h_j^- = 0$, proving the last assertion. $\square$

Example 8.6 (Strict inclusion on the four-torus). On $X = \mathbb{R}^4/\mathbb{Z}^4$, use the Euclidean metric and standard orientation, and put $e^{ij} = dx_i \wedge dx_j$. Set

$$U = e^{12} + e^{34}, \quad V_0 = e^{13} - e^{24}, \quad W = e^{14} + e^{23},$$

$$f = \frac{1}{4} \sin(2\pi x_1), \quad k = \frac{1}{4} \sin(2\pi x_2), \quad r = (1 + f^2 + k^2)^{1/2}.$$

The form $F = (U + fV_0 + kW)/r$ is self-dual and has squared norm two, so it is the fundamental form of a smooth orthogonal almost complex structure J of this orientation. The form

$$\eta = U + 2fe^{13} + 2ke^{23}$$

is closed. Its self-dual part is rF , so it is invariant, and its anti-self-dual part is

$$f(e^{13} + e^{24}) + k(e^{23} - e^{14}).$$

Its Hermitian eigenvalues are $r \pm \sqrt{f^2 + k^2} > 0$. Thus η is compatible. The sine terms are exact, giving $[\eta] = [U]$.

A closed anti-invariant form is self-dual and harmonic for the flat metric, hence is a constant combination $aU + bV_0 + cW$. Orthogonality to F gives $a + bf + ck = 0$ everywhere. Varying x_1, x_2 yields $a = b = c = 0$; thus $H_J^- = 0$. The closed forms $U \pm V_0$ have invariant projections $(1 \pm f)F/r$, so both tame J . Their classes pair to zero, since $U^2 = V_0^2$. Neither class can have a compatible representative, which would wedge strictly positively with a tamer in the other class. Hence

$$\mathcal{K}_J^c + H_J^- = \mathcal{K}_J^c \subsetneq \mathcal{K}_J^t.$$

In Theorem 1.2, the class $[U + V_0]$ fails the strict positivity test against $[U - V_0] \in \mathcal{C}$, and conversely.

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